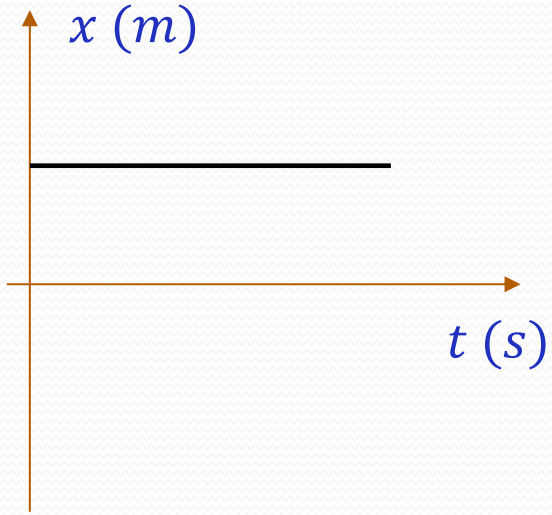
Chapter 2: Motion in a Straight Line
(Constant Acceleration)

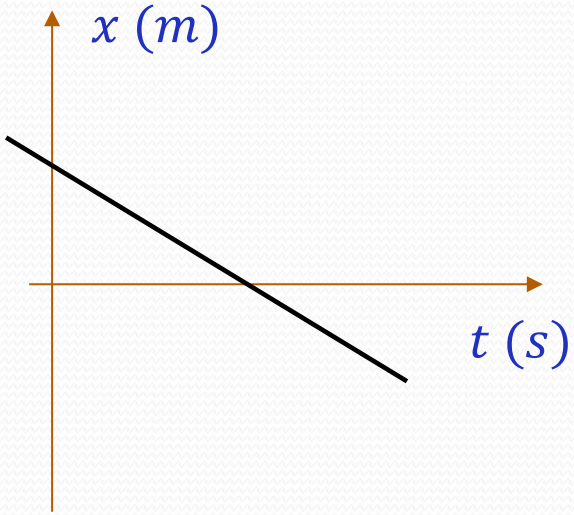
The following subjects will be discussed

- Definitions.
- Equations of Motion.
- A Special Case: Free Fall.
- Examples.

Remarks

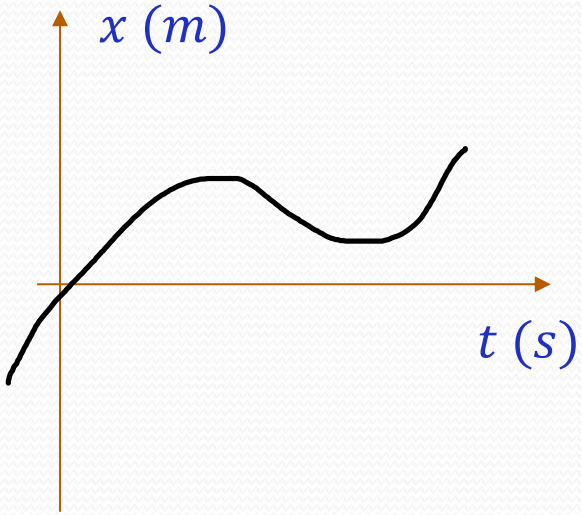


Instantaneous velocity is zero.
Object not moving.



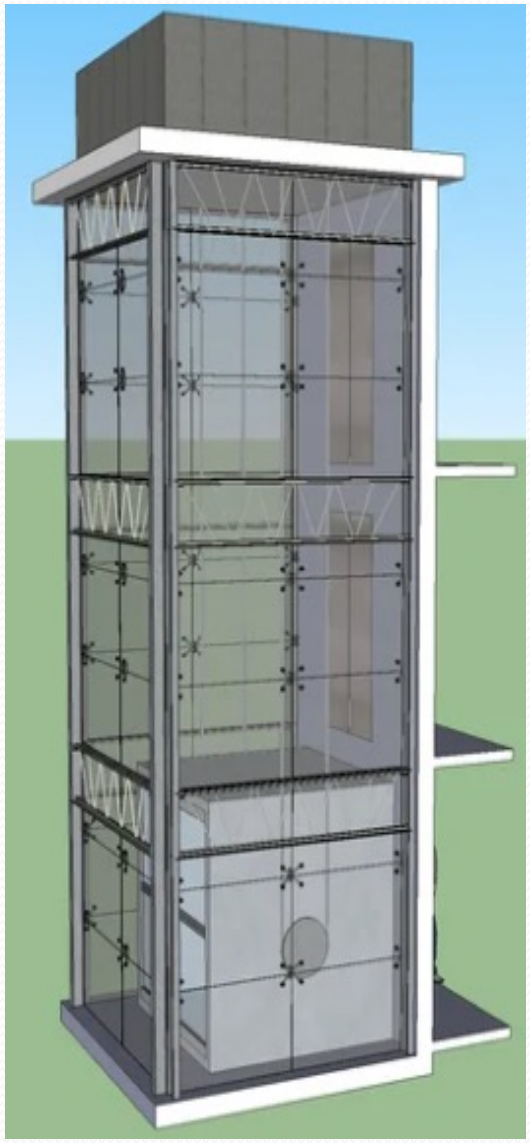
Instantaneous velocity is constant (negative).
Object moving with zero acceleration.

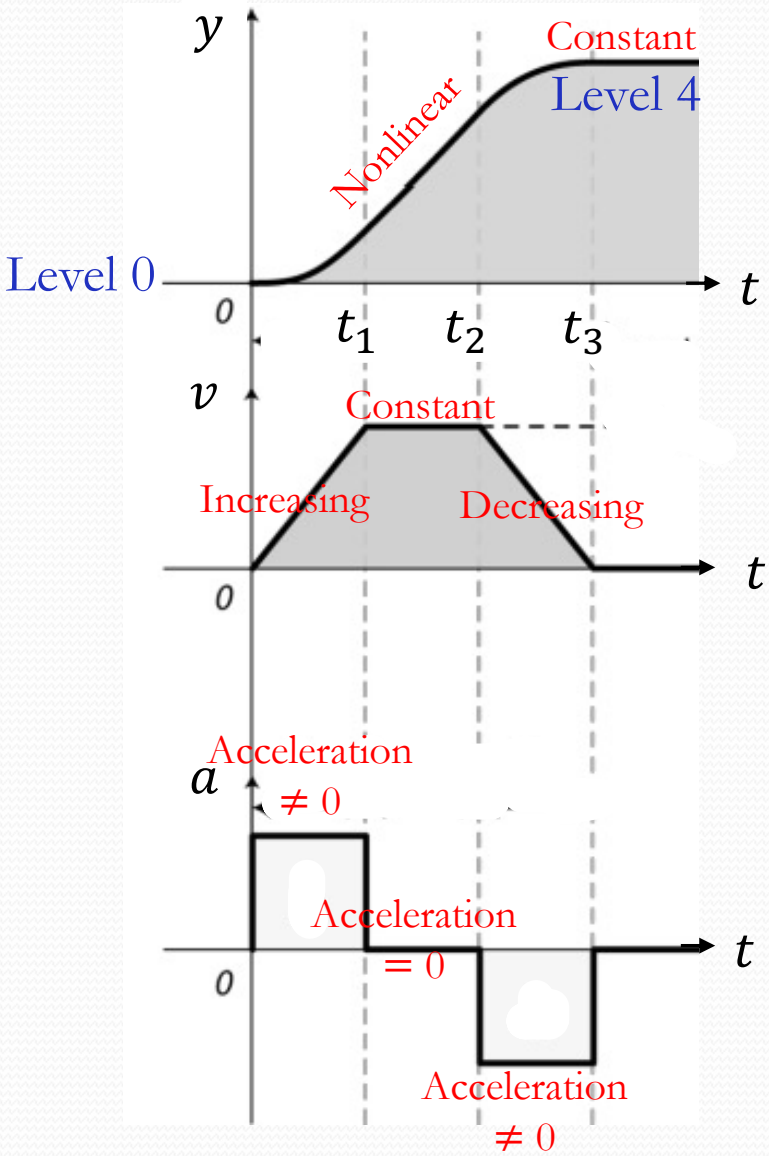
Object moving to the left.



Instantaneous velocity is varying.
Object moving with some acceleration.

The Elevator Example





y (+ve): Elevator at or above ground level.

y (-ve): Elevator at basement levels.

v (+ve): motion at that specific time is Up.

v (-ve): motion at that specific time is Down.

a (+ve): speed at that specific time is increasing.

a (-ve): speed at that specific time is decreasing.

Examples

1) A car is travelling along a straight road for 40 km at 60 km/h forward, then returned on the same road for 30 km at 30 km/h. Find the average velocity and average speed.

$$v_{ave} = \frac{\Delta x}{\Delta t} = \frac{40 - 30}{\frac{40}{60} + \frac{30}{30}} = 5.99 \quad \text{(km/h)}$$

$$Speed_{ave} = \frac{Distance}{\Delta t} = \frac{40 + 30}{\frac{40}{60} + \frac{30}{30}} = 41.9 \quad \text{(km/h)}$$

2) The position of a moving object at any given time is given by $x(t) = 3t - 4t^2 + t^3$

Displacement between t=0 and t=4 s

$x(0) = 0$

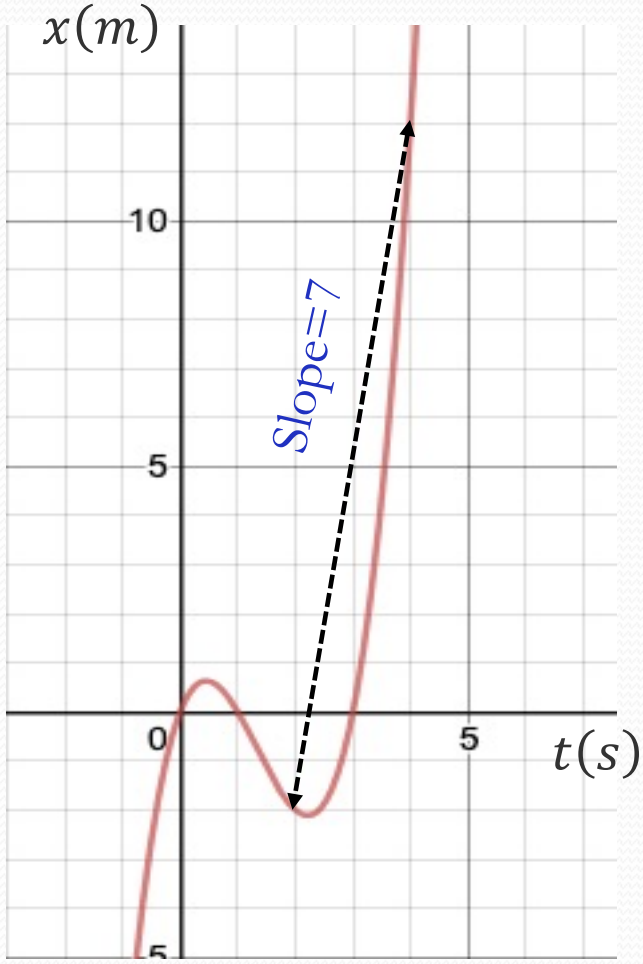
$x(4) = 12$ m

$\Delta x = 12$ m

Average velocity between t=2 and t=4 s

$\Delta x = 12 - (-2) = 14$ m

$v_{ave} = \frac{\Delta x}{\Delta t} = \frac{14}{2} = 7$ m/s



The Equations of Motion

These equations apply for the special case when acceleration is constant.

Because the acceleration is constant, we can write

$$a = a_{ave} = \frac{\Delta v}{\Delta t} = \frac{v - v_o}{t - t_o}$$

$\Rightarrow$ $t_o = 0$

Initial conditions

$$v - v_o = a(t - t_o)$$

$$v(t) = at + v_o$$

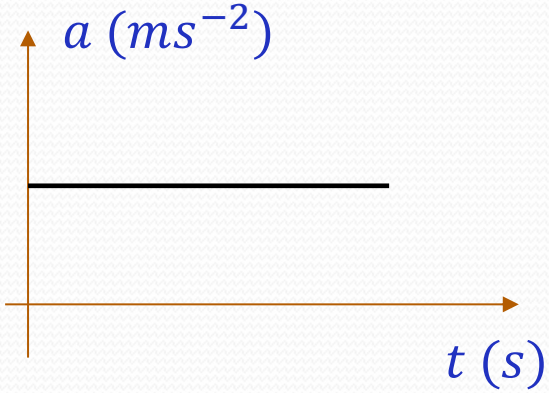
From the definition of the average velocity

$$v_{ave} = \frac{\Delta x}{\Delta t} = \frac{x - x_o}{t - t_o} = \frac{1}{2}(v - v_o)$$

$\Rightarrow$ $t_o = 0$

$$x - x_o = v_o t + \frac{1}{2}at^2$$

$$x(t) = x_o + v_o t + \frac{1}{2}at^2$$



That's it!

The Equations of Motion

$$v(t) = at + v_o \qquad 1$$

$$x(t) = x_o + v_o t + \frac{1}{2}at^2 \qquad 2$$

$$v^2(t) = {v_o}^2 + 2a \cdot [x(t) - x_o] \qquad 3$$

$$x(t) = x_o + \frac{1}{2}[v(t) + v_o] \cdot t \qquad 4$$

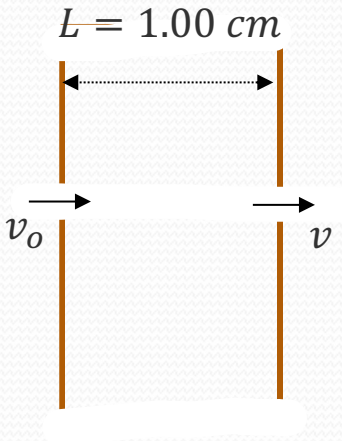
$$x(t) = x_o + v(t) \cdot t - \frac{1}{2}at^2 \qquad 5$$

Examples

1) An electron enters a region (L=1.00 cm) of constant electric field with initial velocity of 1.50×10^5 m/s. It accelerates to a velocity of 5.70×10^6 m/s when it exits that region. Find the acceleration.

Which equation?

$$v^2(t) = {v_o}^2 + 2a\cdot[x(t) - x_o]$$



$$v(t) = at + v_o \quad 1$$

$$x(t) = x_o + v_o t + \frac{1}{2}at^2 \quad 2$$

$$v^2(t) = {v_o}^2 + 2a\cdot[x(t) - x_o] \quad 3$$

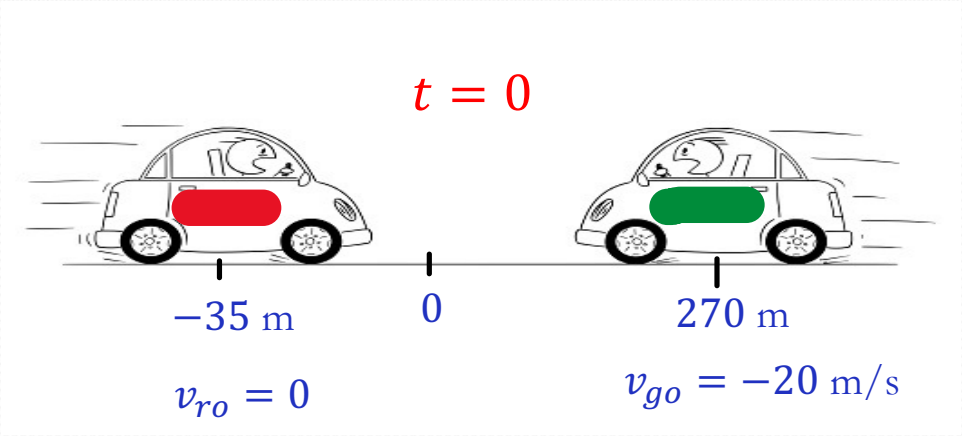
$$x(t) = x_o + \frac{1}{2}[v(t) + v_o]\cdot t \quad 4$$

$$x(t) = x_o + v(t)\cdot t - \frac{1}{2}at^2 \quad 5$$

$$a = \frac{v^2 - {v_o}^2}{2(x - x_o)} = \frac{(5.70\times10^6)^2 - (1.50\times10^5)^2}{2(1.00\times10^{-2})} = 1.62\times10^{15}\ (\mathrm{m/s^2})$$

Examples

2) These two cars have regrettably collided head-on at t=12 s. Find the acceleration of the red car.



For the green car,

$$a_g = 0$$

$$x(t) = x_o + v_o t + \frac{1}{2}at^2$$

$$x = x_o + v_o t$$

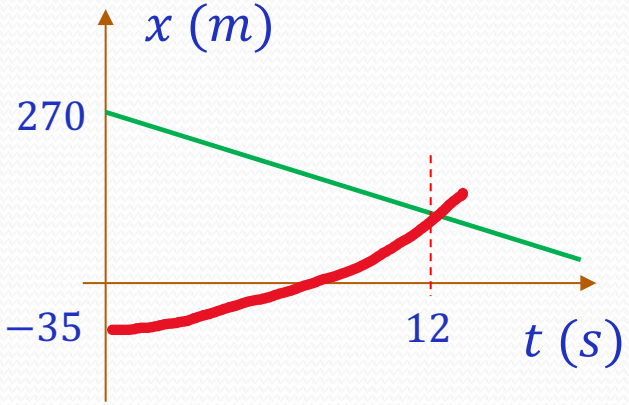
$$x = 270 + (-20)(12) = 30 \text{ (m)}$$

For the red car,

$$x(t) = x_o + v_o t + \frac{1}{2}at^2$$

$$x = -35 + \frac{1}{2}a_r(12)^2 = 30 \text{ (m)}$$

$$a_r = 0.9 \quad (\text{m/s}^2)$$



Free-fall Analysis

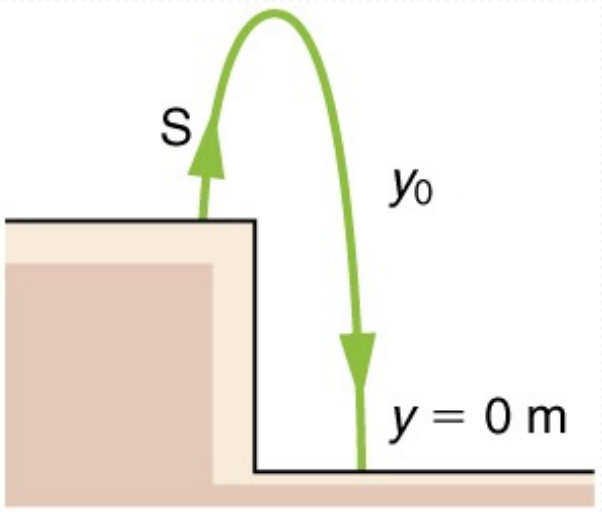
Object falling under the effect of gravity only (constant gravity acceleration g).

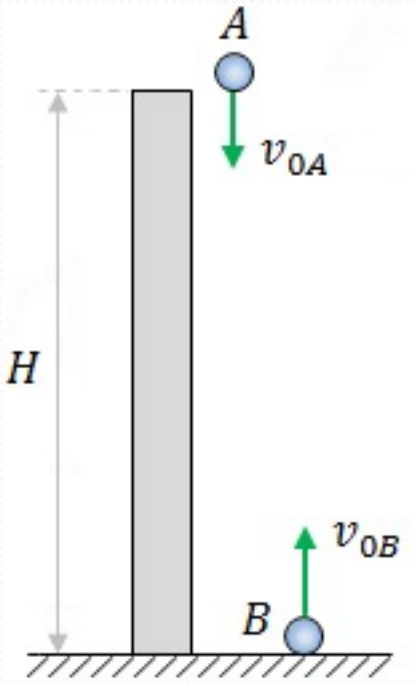
Ignore all other effects (forces).

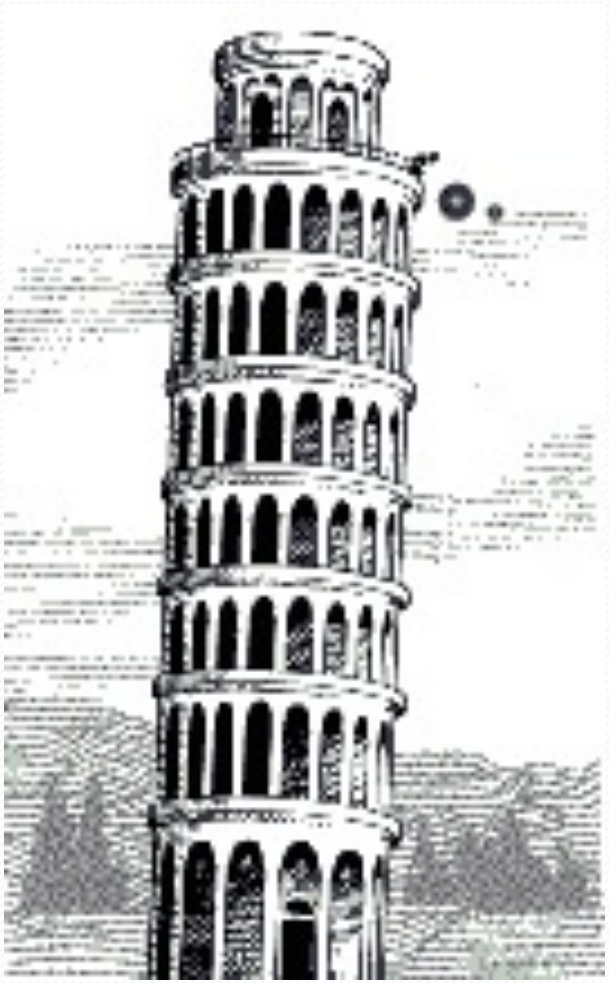
All equations of motion apply with $a = -g$

Acceleration due to gravity $g = 9.8$ (m/s^2)

Motion is vertical (use y)







Free-fall Equations

$$v(t) = v_o - gt \qquad 1$$

$$y(t) = y_o + v_o t - \frac{1}{2} g t^2 \qquad 2$$

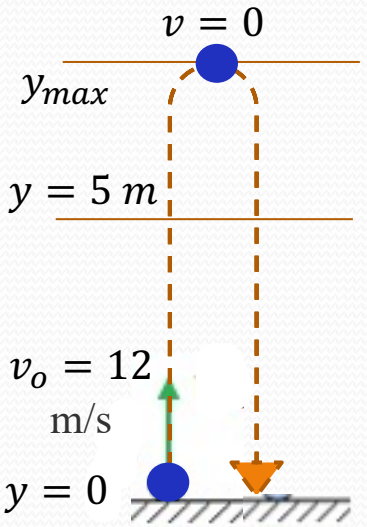
$$v^2(t) = {v_o}^2 - 2g \cdot [y(t) - y_o] \qquad 3$$

$$y(t) = y_o + \frac{1}{2} [v(t) + v_o] \cdot t \qquad 4$$

$$y(t) = y_o + v(t) \cdot t + \frac{1}{2} g t^2 \qquad 5$$

Examples

1) The object is thrown vertically upward with initial velocity of 12 m/s. Find the roundtrip time, maximum height and the time to reach 5 m.



Going up time

$$v(t) = v_o - gt \Rightarrow 0 = 12 - 9.8\,t \Rightarrow t = 1.2\ s$$

$$y(t) = y_o + v_o t - \frac{1}{2}gt^2 \Rightarrow y_{max} = 0 + 12\,(1.2) - \frac{1}{2}(9.8)(1.2)^2$$

$$y_{max} = 7.34\ \text{m}$$

Going down

$$y(t) = y_o + v_o t - \frac{1}{2}gt^2 \Rightarrow 0 = 7.34 + (0)\,t - \frac{1}{2}9.8t^2$$

$$t = 1.2\ s \Rightarrow t_{total} = 2.4\ s$$

Velocity hitting the ground??

Time to reach 5 m

$$y(t) = y_o + v_o t - \frac{1}{2}gt^2 \Rightarrow 5 = 0 + 12\,t - \frac{1}{2}9.8t^2$$

$$t = 0.53\ s \qquad t = 1.9\ s$$

$$v(t) = v_o - gt \quad 1$$

$$y(t) = y_o + v_o t - \frac{1}{2}gt^2 \quad 2$$

$$v^2(t) = v_o{}^2 - 2g \cdot [y(t) - y_o] \quad 3$$

$$y(t) = y_o + \frac{1}{2}[v(t) + v_o] \cdot t \quad 4$$

$$y(t) = y_o + v(t) \cdot t + \frac{1}{2}gt^2 \quad 5$$

Examples

2) A balloon heading upwards at a constant speed of 19.6 m/s losses some of its cargo as given by the curve below. At what height did this incident happen?

Will go higher for 2 s

$$y(t) - y_o = v_o t - \frac{1}{2} g t^2$$

$$d = v_o t - \frac{1}{2} g t^2$$

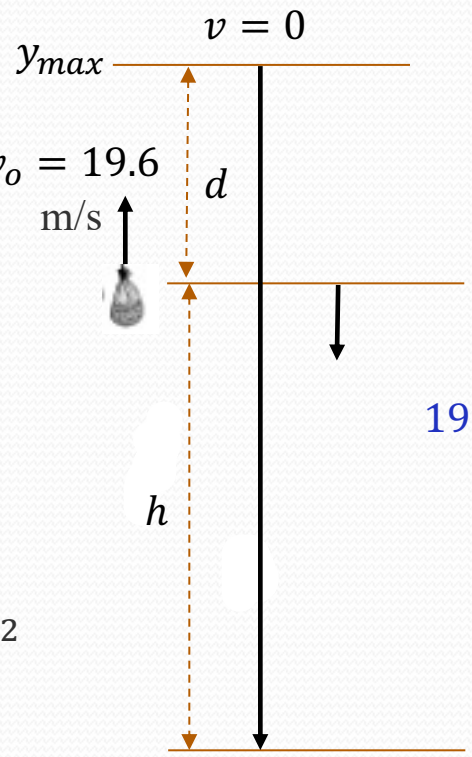
$$d = (19.6)(2) - \frac{1}{2}(9.8)(2)^2$$

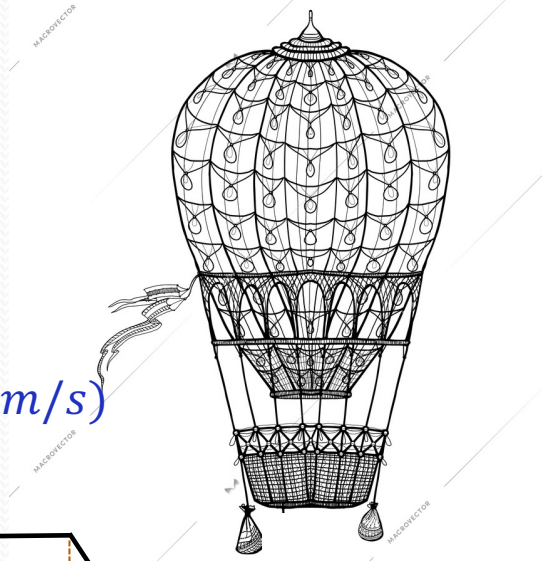
$$d = 19.6 \text{ m}$$

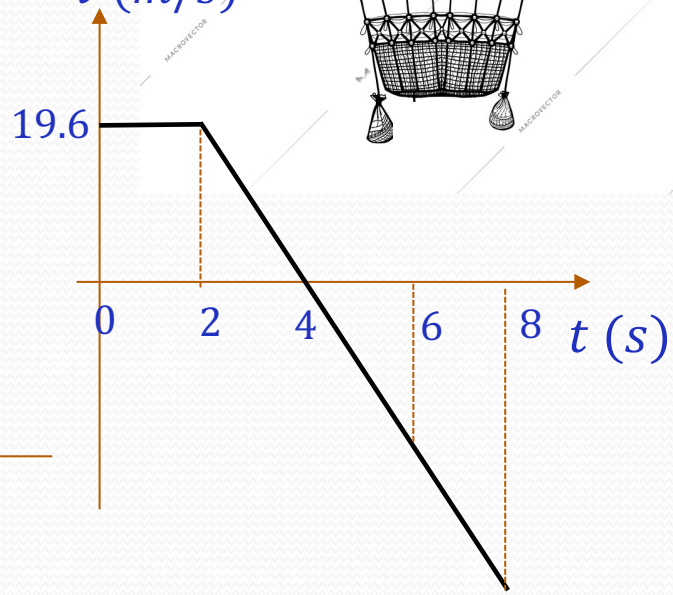
$$y(t) - y_o = v_o t - \frac{1}{2} g t^2$$

$$0 - h = (-19.6)(2) - \frac{1}{2}(9.8)(2)^2$$

$$h = 58.8 \text{ m}$$







Chapter 3: Vector Analysis

The following subjects will be discussed

- Vector Representation.
- Basic Vector Operations.
- Examples.

Vector Representation

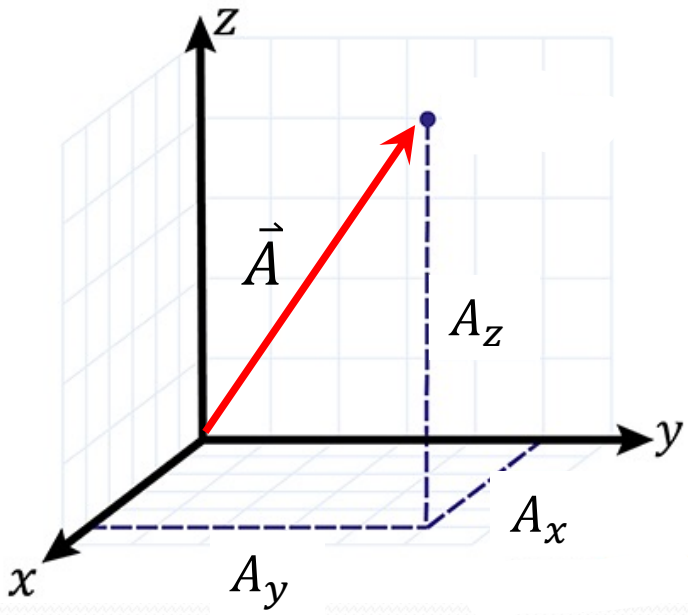
Physical quantities are either scalar or vector quantities.

Scalar quantities have no sense of direction: mass, temperature, volume, power, etc.

Vector quantities have a sense of direction. They are defined by magnitude and direction: position, velocity, acceleration, torque, momentum, force, etc.

In 1D, a simple sign (plus or minus) is enough to indicate direction (left, right, etc.)

In 2D and 3D, a special mathematical format is used (instead of multiple signs).



A vector quantity $\vec{A}$ is generally represented as

$$\vec{A} = A_x\,\vec{\imath} + A_y\,\vec{\jmath} + A_z\,\vec{k}$$

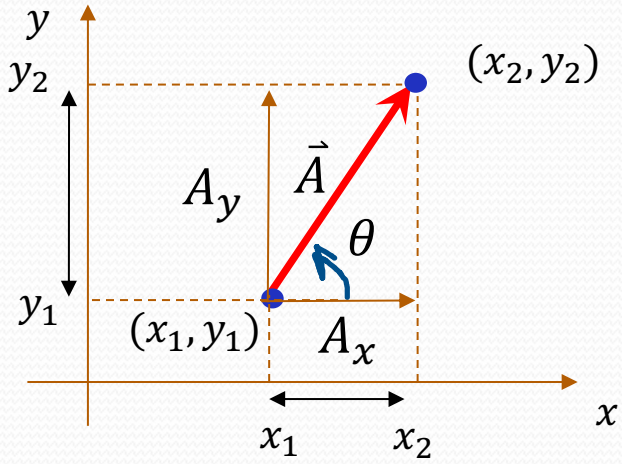
The vector Magnitude $|\vec{A}| = \sqrt{{A_x}^2 + {A_y}^2 + {A_z}^2}$

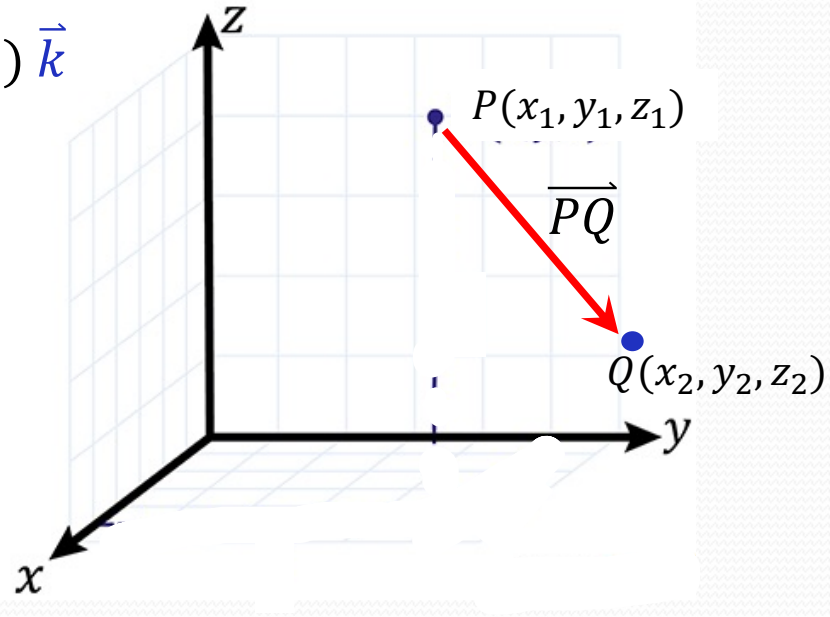
The direction is given by the "unit vector" $\vec{a}_A = \dfrac{\vec{A}}{|A|}$ $\vec{A} = |A|\vec{a}_A$

Vector Representation

Given two points P and Q, the vector $\overrightarrow{PQ}$ is given by

$$\overrightarrow{PQ} = (x_2 - x_1)\,\vec{\imath} + (y_2 - y_1)\,\vec{\jmath} + (z_2 - z_1)\,\vec{k}$$





$$\vec{A} = (x_2 - x_1)\,\vec{\imath} + (y_2 - y_1)\,\vec{\jmath}$$

$$\vec{A} = A_x\,\vec{\imath} + A_y\,\vec{\jmath}$$

$$|\vec{A}| = \sqrt{{A_x}^2 + {A_y}^2}$$

$$A_x = |A| \cos\theta$$

$$A_y = |A| \sin\theta$$

$$\theta = \tan^{-1}\frac{A_y}{A_x}$$

Position and Displacement Vectors

The position vector of P is

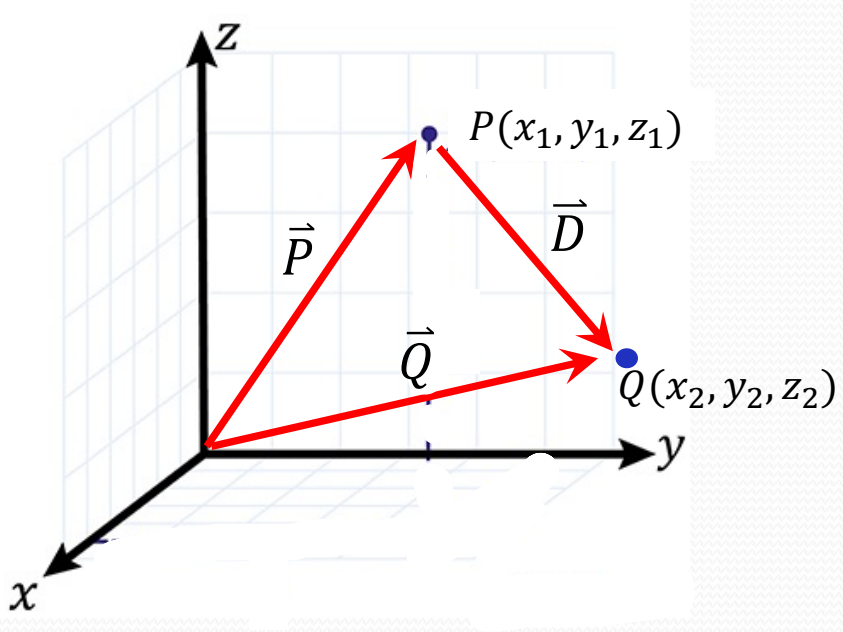
$$\vec{P} = x_1\,\vec{i} + y_1\,\vec{j} + z_1\,\vec{k}$$

The position vector of Q is

$$\vec{Q} = x_2\,\vec{i} + y_2\,\vec{j} + z_2\,\vec{k}$$

The displacement vector $\overrightarrow{PQ}$ is

$$\overrightarrow{PQ} = (x_2 - x_1)\,\vec{i} + (y_2 - y_1)\,\vec{j} + (z_2 - z_1)\,\vec{k}$$



The length (magnitude) of the vector represents the distance.

Examples

Given ***P*** (0, 2, 4) and ***Q*** (-3, 1, 5),

The position vector of P is $\vec{P} = 0\,\vec{\imath} + 2\,\vec{\jmath} + 4\,\vec{k}$

The position vector of Q is $\vec{Q} = -3\,\vec{\imath} + 1\,\vec{\jmath} + 5\,\vec{k}$

The displacement vector $\overrightarrow{PQ}$ is $\overrightarrow{PQ} = (-3-0)\,\vec{\imath} + (1-2)\,\vec{\jmath} + (5-4)\,\vec{k}$

$$\overrightarrow{PQ} = -3\,\vec{\imath} - \vec{\jmath} + \vec{k}$$

The distance from P to Q is $\left|\overrightarrow{PQ}\right| = \sqrt{(-3)^2+(1)^2+(1)^2} = \sqrt{11}$

A unit vector along $\overrightarrow{PQ}$ is $\vec{a}_{PQ} = \dfrac{\overrightarrow{PQ}}{\left|\overrightarrow{PQ}\right|}$

$$\vec{a}_{PQ} = \frac{-3}{\sqrt{11}}\,\vec{\imath} - \frac{1}{\sqrt{11}}\,\vec{\jmath} + \frac{1}{\sqrt{11}}\,\vec{k}$$

Basic Vector Operations

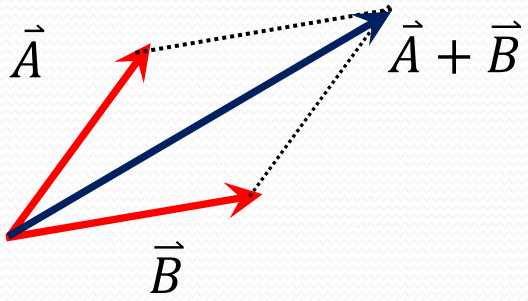
Addition/subtraction

$$\vec{A} \pm \vec{B} = (A_x \pm B_x)\,\vec{\imath} + (A_y \pm B_y)\,\vec{\jmath} + (A_z \pm B_z)\,\vec{k}$$

$$\vec{A} \pm \vec{B} = \vec{B} \pm \vec{A}$$

$$\vec{A} + \vec{B} + \vec{C} = (\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$

$$k(\vec{A} \pm \vec{B}) = k\vec{A} \pm k\vec{B}$$



Example

Given $\vec{A} = 10\,\vec{\imath} - 4\,\vec{\jmath} + 6\,\vec{k}$ $\vec{B} = 2\,\vec{\imath} + \vec{\jmath}$

1) The component of $\vec{A}$ along the y-direction -4

2) The magnitude of $3\vec{A} - \vec{B}$

$$3\vec{A} - \vec{B} = 30\,\vec{\imath} - 12\,\vec{\jmath} + 18\,\vec{k} - 2\,\vec{\imath} - \vec{\jmath} = 28\,\vec{\imath} - 13\,\vec{\jmath} + 18\,\vec{k}$$

$$\left|3\vec{A} - \vec{B}\right| = \sqrt{(28)^2 + (13)^2 + (18)^2} = \sqrt{1277}$$

Basic Vector Operations

Example

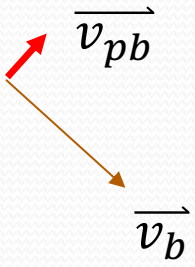
$$\overrightarrow{v_b} = 10(\cos(-45)\ \vec{i} + \sin(-45)\ \vec{j}\,)$$

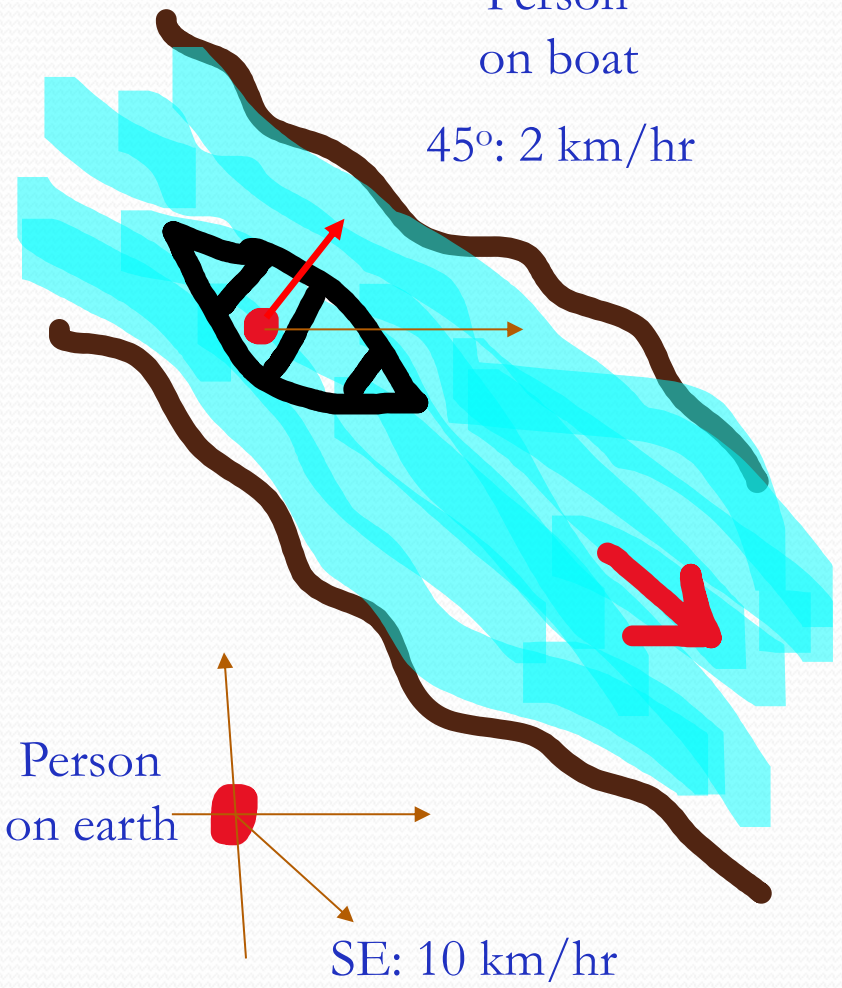
$$\overrightarrow{v_{pb}} = 2(\cos(45)\ \vec{i} + \sin(45)\ \vec{j}\,)$$

$$\overrightarrow{v_p} = \overrightarrow{v_b} + \overrightarrow{v_{pb}}$$

$$\overrightarrow{v_p} = 8.48\ \vec{i} - 5.66\vec{j} \quad \text{km/hr}$$

$$\overrightarrow{v_p} = 10.2 \angle -33.72^{o} \quad \text{km/hr}$$





Basic Vector Operations

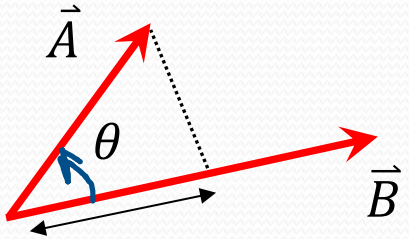
Scalar Multiplication (dot product)

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos\theta$$

$$\vec{A} \cdot \vec{B} = (A_x \cdot B_x) + (A_y \cdot B_y) + (A_z \cdot B_z)$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

Scalar result



Shadow (projection) of $\vec{A}$ on $\vec{B}$

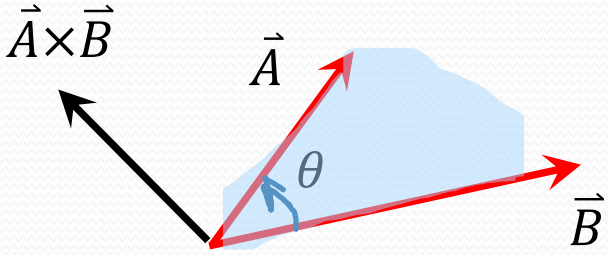
$|\vec{A}|\cos\theta$

Vector Multiplication (cross product)

$$\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta\ \overrightarrow{a_n}$$

Vector result

$$\vec{A}\times\vec{B} = \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$



$$\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)\vec{\imath} - (A_xB_z - A_zB_x)\vec{\jmath} + (A_xB_y - A_yB_x)\vec{k}$$

$$\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$$

Basic Vector Operations

Example

Given $\vec{A} = 3\,\vec{\imath} + 4\,\vec{\jmath} + \vec{k}$ $\qquad \vec{B} = 2\,\vec{\jmath} - 5\,\vec{k}$

Find θ_{AB}.

$$\vec{A} \cdot \vec{B} = (3 \cdot 0) + (4 \cdot 2) + (1 \cdot -5)$$

$$\vec{A} \cdot \vec{B} = 3$$

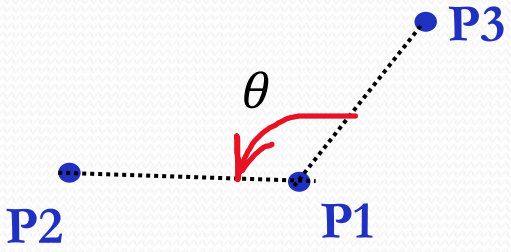
$$|\vec{A}| = \sqrt{26} \qquad |\vec{B}| = \sqrt{29}$$

$$\cos\theta_{AB} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} \qquad \theta_{AB} = 83.7°$$

Basic Vector Operations

Example Show that the points **P1** (5, 2, -4), **P2** (1, 1, 2) and **P3** (-3, 0, 8) all lie on a straight line.

Let's check the angle θ. If $\theta = 180°$, the three points lie on a straight line



$$\overrightarrow{P_1P_2} = -4\,\vec{\imath} - \vec{\jmath} + 6\,\vec{k} \qquad \left|\overrightarrow{P_1P_2}\right| = \sqrt{53}$$

$$\overrightarrow{P_1P_3} = -8\,\vec{\imath} - 2\,\vec{\jmath} + 12\,\vec{k} \qquad \left|\overrightarrow{P_1P_3}\right| = \sqrt{212}$$

$$\overrightarrow{P_1P_2} \cdot \overrightarrow{P_1P_3} = 32 + 2 + 72 = 106$$

$$\vec{A} \cdot \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\cos\theta \quad \Rightarrow \quad \cos\theta = \frac{\vec{A} \cdot \vec{B}}{\left|\vec{A}\right|\left|\vec{B}\right|} = 1 \quad \Rightarrow \quad \theta = 0, 180°$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ -4 & -1 & 6 \\ -8 & -2 & 12 \end{vmatrix} = 0$$

Q & A

Send me an email if
you have any
questions.